

Behaviour of Hub device

- Connects multiple devices in LAN.
- When it receives data on one port, it broadcasts it to all other ports.

→ Layer 1 device

- If two or more devices try to send data at same time, Collision occurs.
- Use Half-duplex Communication.

Behaviour of switch

- Intelligent device used to connect devices in LAN.

→ Layer 2 device

- Contains MAC table (Physical Port ^{No.} vs Source MAC address)
- Performs address learning
- When it receives data on port, it reads the MAC address of sender device and saves in table. (Time validity: 5 mins)
- Initially it broadcast data to all ports, to find MAC address of destination device, then further transmission is forward only to the port of destination device (unicast)
- Full Duplex Communication
- No collision (Dedicated physical link between all ports)

Behaviour of Router:

- Break single broadcast domain
- Layer 3 device.
- Will do routing
- Has a routing table (contains list of routes)
- used to connect different networks.
- Checks the destination IP ~~is~~ has route in the routing table, then it unicast to the corresponding network. if route is not there, discard the data.

Routing table → Network ID / Prefix length - Interface number